Physics-Informed Neural Network for a Reduced WNT-HOX-Retinoic Acid Model of Colorectal Cancer

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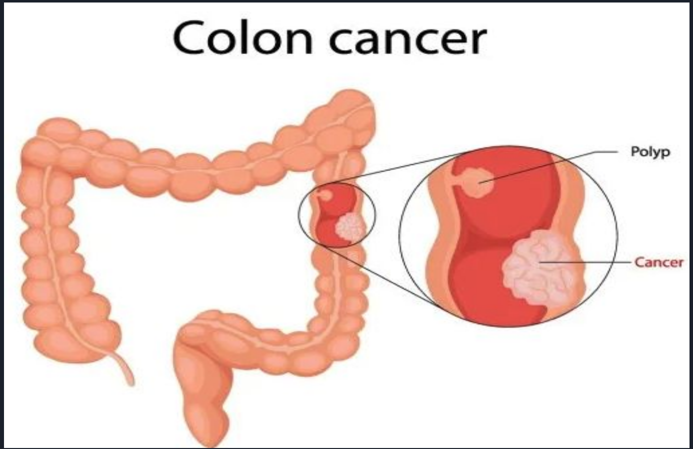

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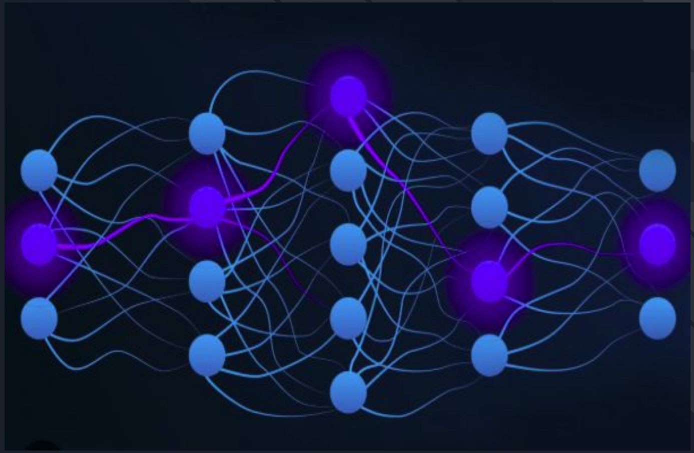


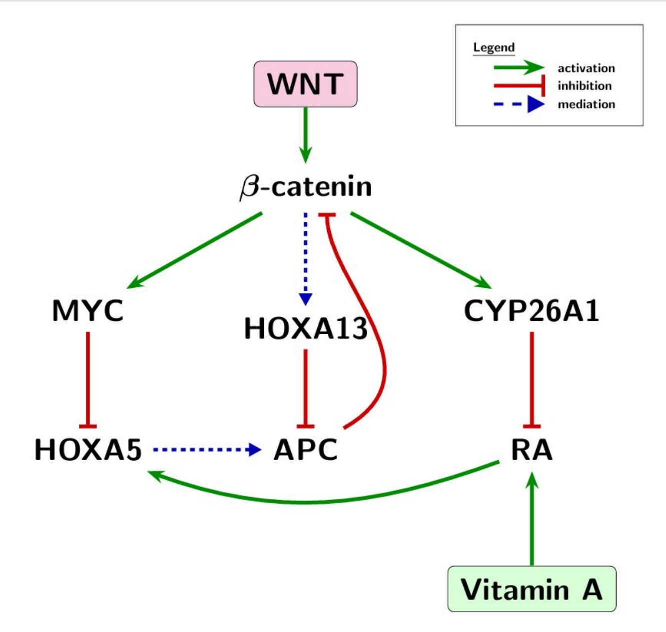
Overview

- Biology/Schematic
- Equations
- Forward PINN
- Inverse PINN
- Inverse Bayesian PINN
- Next Steps

Biological Network Governing WNT–HOX–RA Signaling

Objective

Develop a reduced mathematical model describing the interactions among the **WNT**, **HOX**, and **Retinoic Acid (RA)** pathways during colorectal cancer progression.



State Variables

$y = [B(t), P(t), H_5(t), H_{13}(t), M(t), R(t), C(t)]$

B = β-catenin

P = APC

H_5 = HOXA5

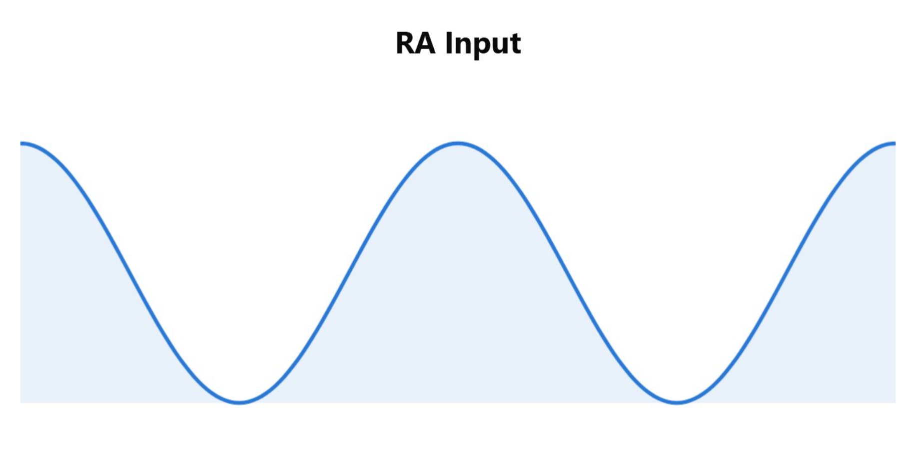
H_{13} = HOXA13

M = MYC

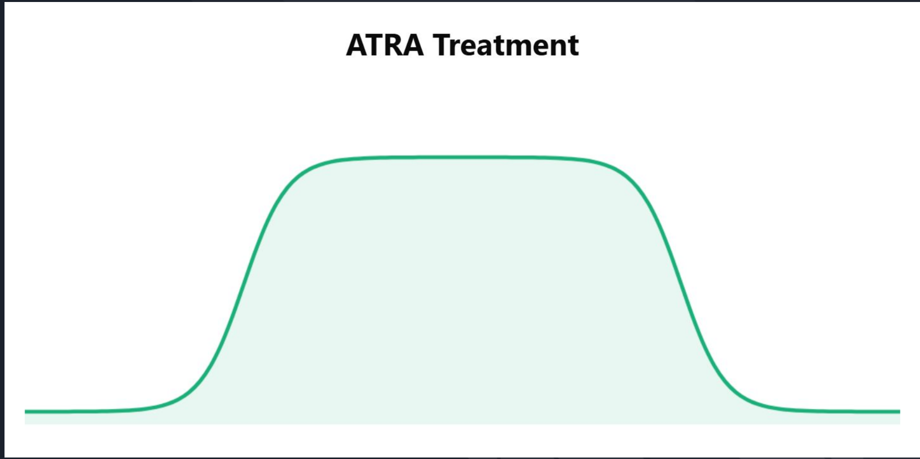
R = Retinoic Acid

C = CYP26A1

RA Input + Treatment



$$A_d \left[1 + \cos\left(\frac{2\pi t}{T_d} - \phi \right) \right]$$



$$u_R(t) = u_0 + \frac{D}{2} \left[\tanh(q(t - t_1)) - \tanh(q(t - t_2)) \right].$$

Dimensional Model

$$\frac{dB}{dt} = a_B W + v_{13}\frac{H_{13}^{n_H}}{K_{13}^{n_H} + H_{13}^{n_H}} - d_B B - k_{PB} P B - v_5 \frac{H_5 B}{K_5 + B},$$

$$\frac{dP}{dt} = a_P \frac{1 + \rho_5 H_5}{1 + \rho_B B + \rho_{13} H_{13}} - \Big(d_{P0} + d_{P1}(1 - \Theta_P)\Big) P,$$

$$\frac{dH_5}{dt} = a_5 + v_R \frac{R}{K_R + R} - d_5 H_5 - v_M \frac{M H_5}{K_M + M},$$

$$\frac{dH_{13}}{dt} = a_{13} + v_{B13}\frac{B^{n_B}}{K_{B13}^{n_B} + B^{n_B}} + v_{M13}\frac{M^{n_M}}{K_{M13}^{n_M} + M^{n_M}} - d_{13} H_{13},$$

$$\frac{dM}{dt} = a_M + v_{BM}\frac{B^{n_B}}{K_{BM}^{n_B} + B^{n_B}} - d_M M,$$

$$\frac{dR}{dt} = u_0 + A_d\left[1 + \cos\left(\frac{2\pi t}{T_d} - \phi\right)\right] + \frac{D}{2}\Big[\tanh(q(t - t_1)) - \tanh(q(t - t_2))\Big] - d_R R - k_{CR} C R,$$

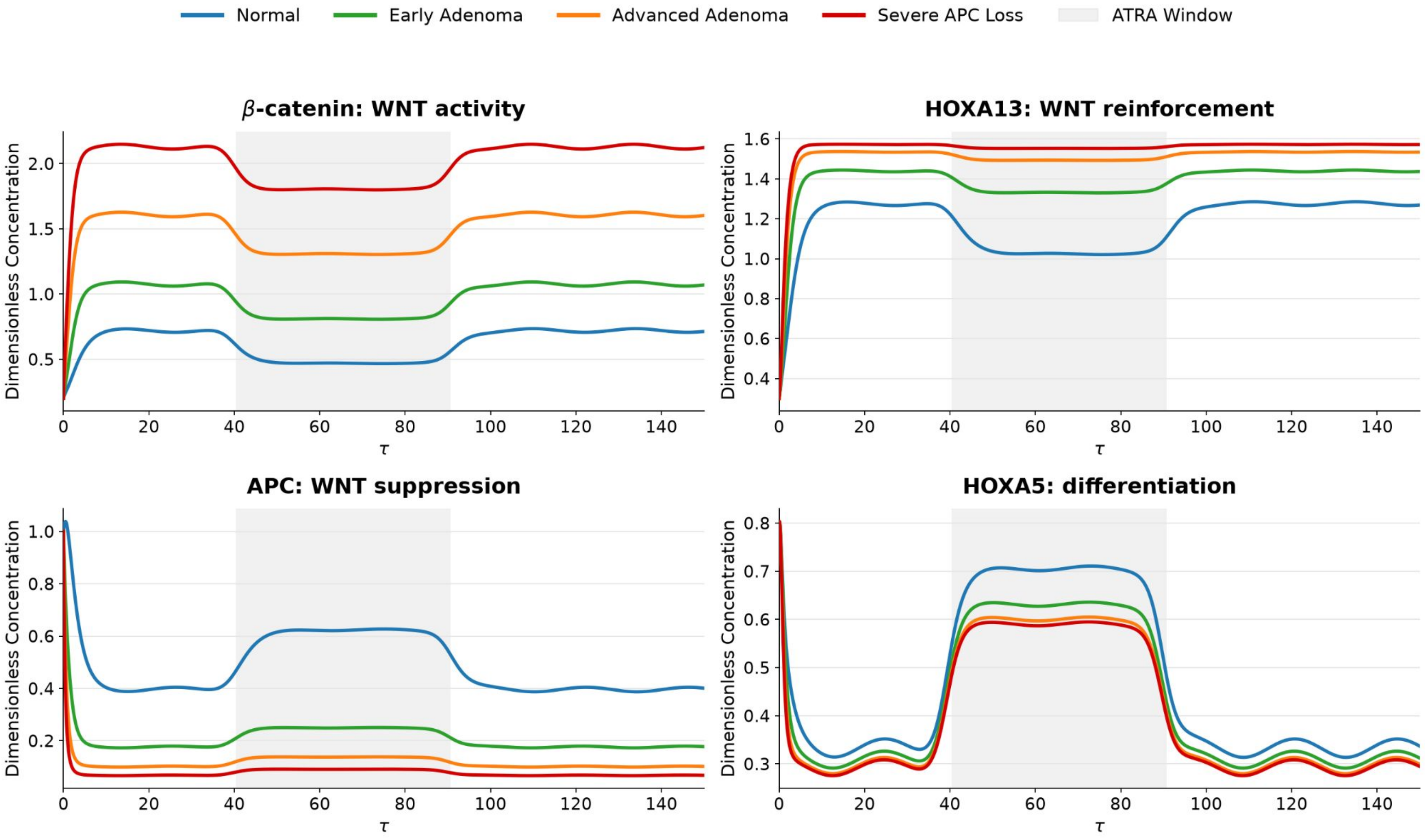
$$\frac{dC}{dt} = a_C + v_{RC}\frac{R}{K_{RC} + R} + v_{BC}\frac{B^{n_B}}{K_{BC}^{n_B} + B^{n_B}} - d_C C.$$

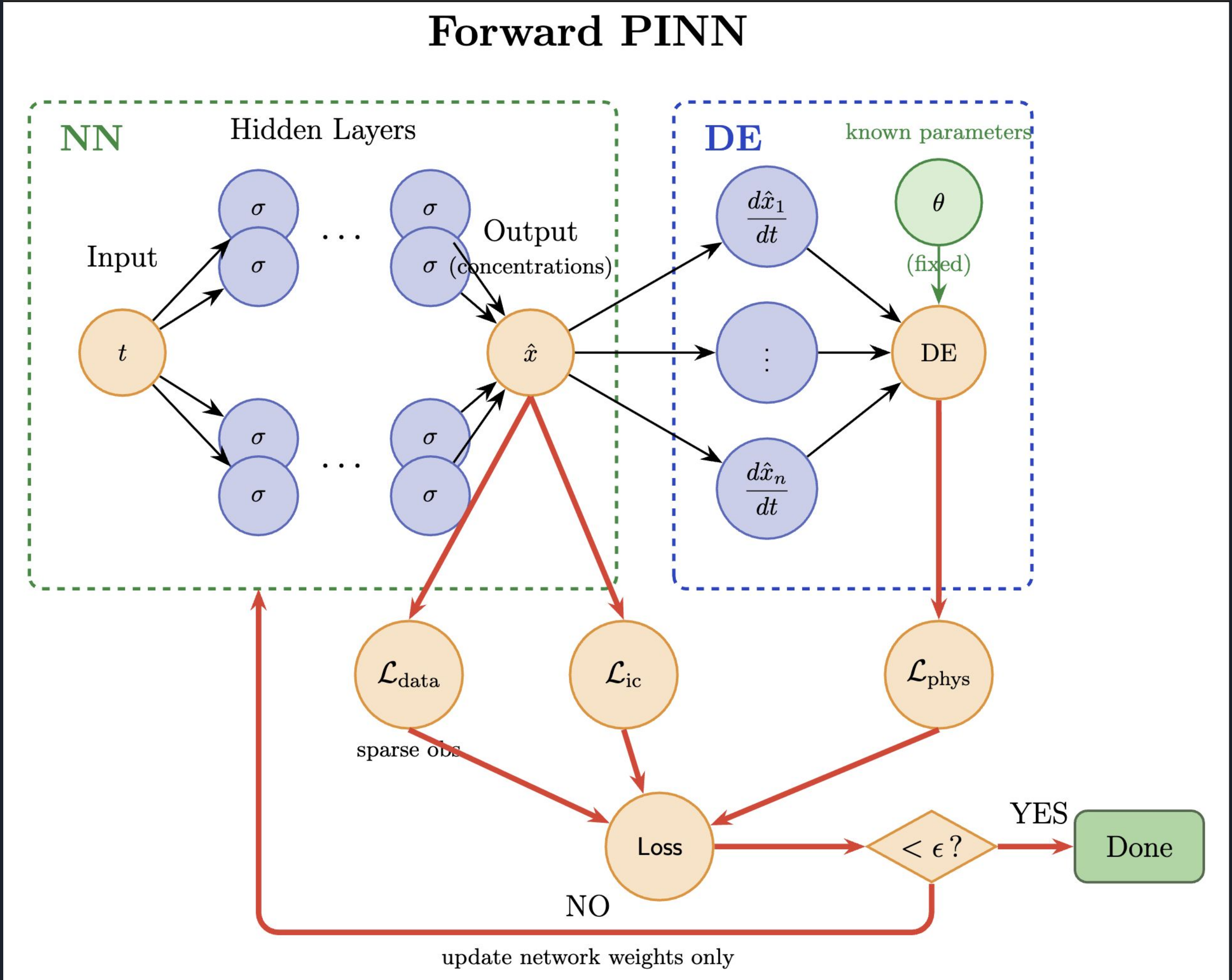
4 Regimes

Disease progression is modeled by increasing WNT signaling (W) and progressive loss of APC function (Θ_p), mimicking CRC progression.

- "Normal"
  - W = **0.8**, Θ_p = **1.0**
- "Early Adenoma"
  - W = **1.0**, Θ_p = **0.75**
- "Advanced Adenoma"
  - W = **1.5**, Θ_p = **0.5**
- "Severe APC Loss"
  - W = **2.0**, Θ_p = **0.25**

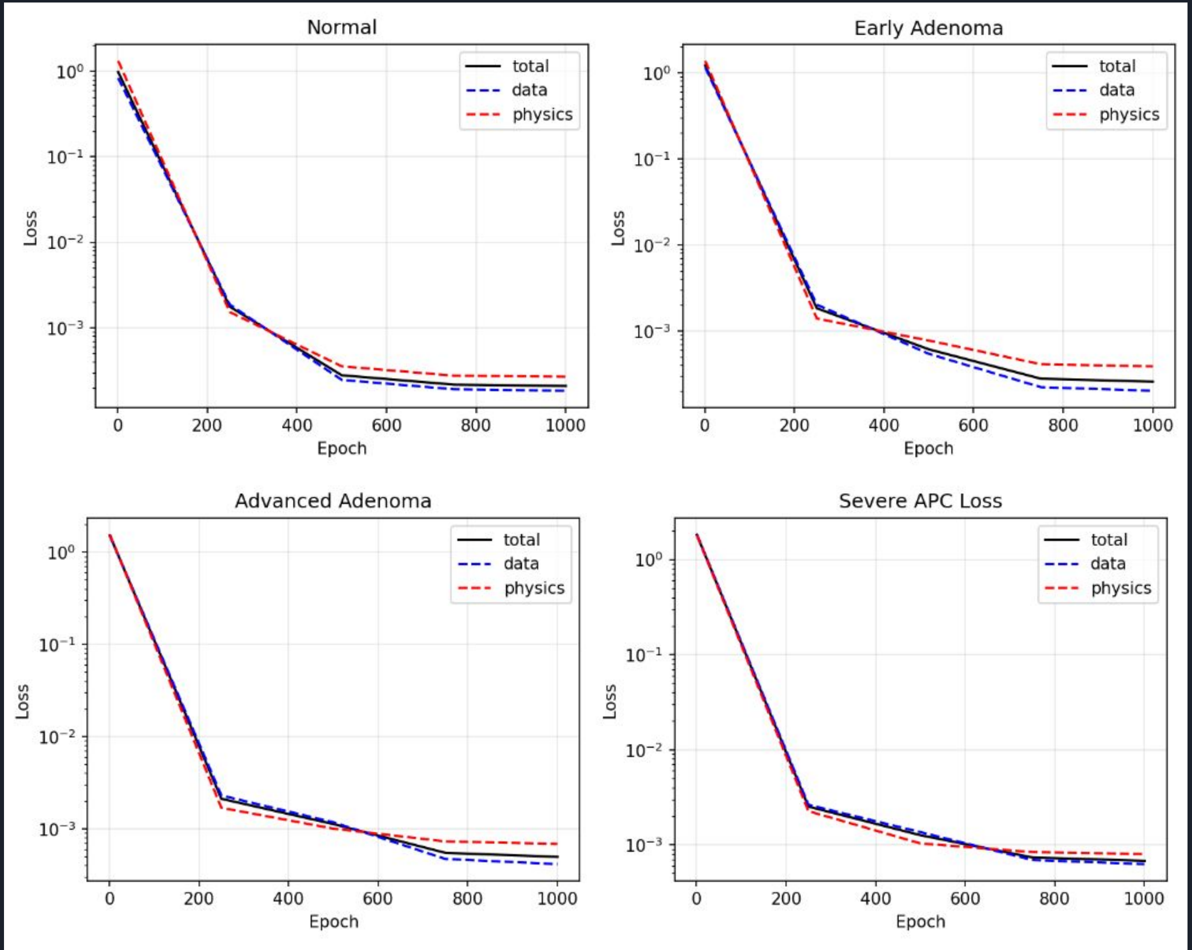
Core WNT-HOX Regulatory Dynamics

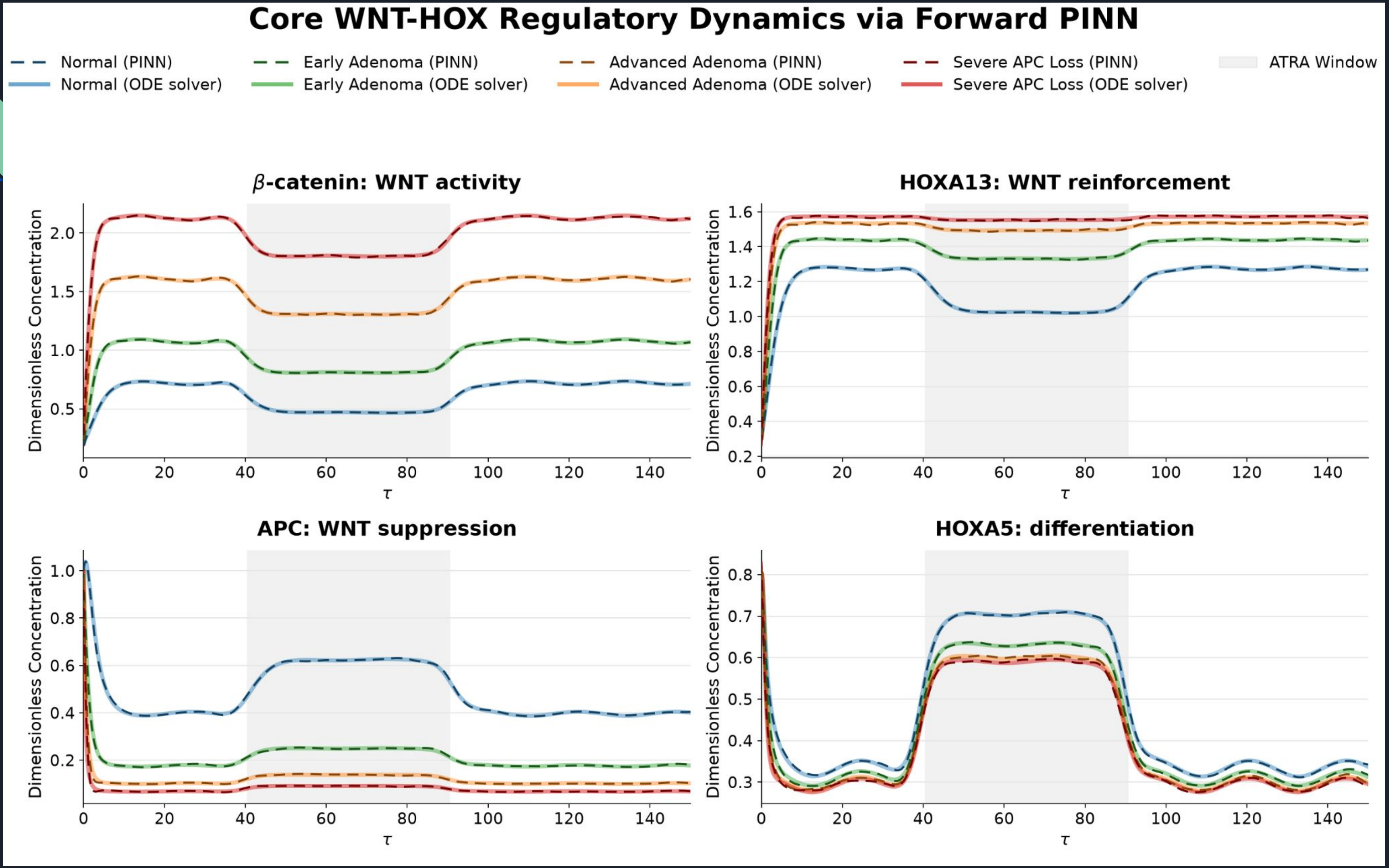


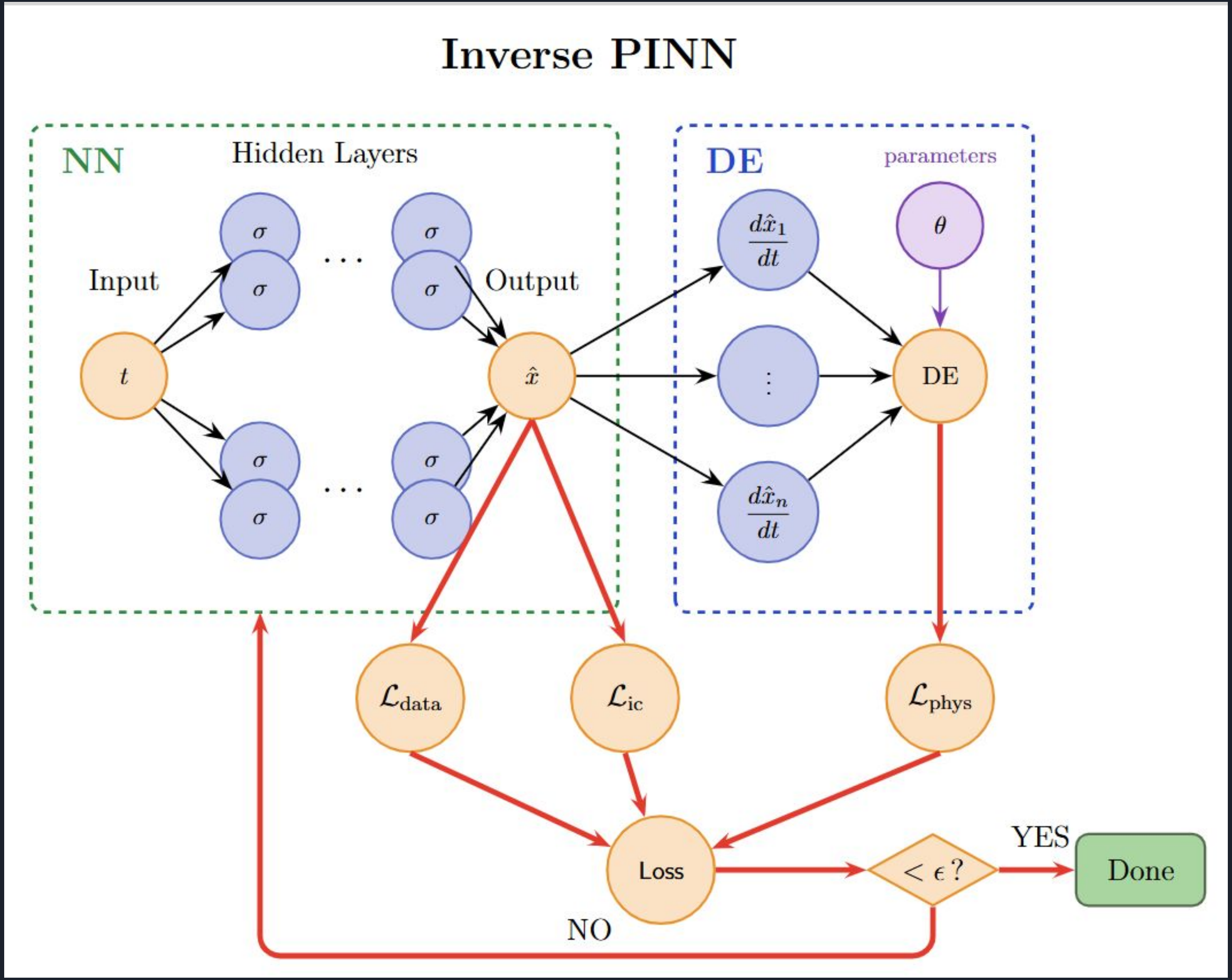


Forward PINN
NN
Hidden Layers
Input
t
σ
...
Output
(concentrations)
$\hat{x}$
DE
known parameters
θ
(fixed)
$\frac{d\hat{x}_1}{dt}$
$\vdots$
$\frac{d\hat{x}_n}{dt}$
DE
$\mathcal{L}_{\text{data}}$
sparse obs
$\mathcal{L}_{\text{ic}}$
$\mathcal{L}_{\text{phys}}$
Loss
$< \epsilon\,?$
YES
Done
NO
update network weights only

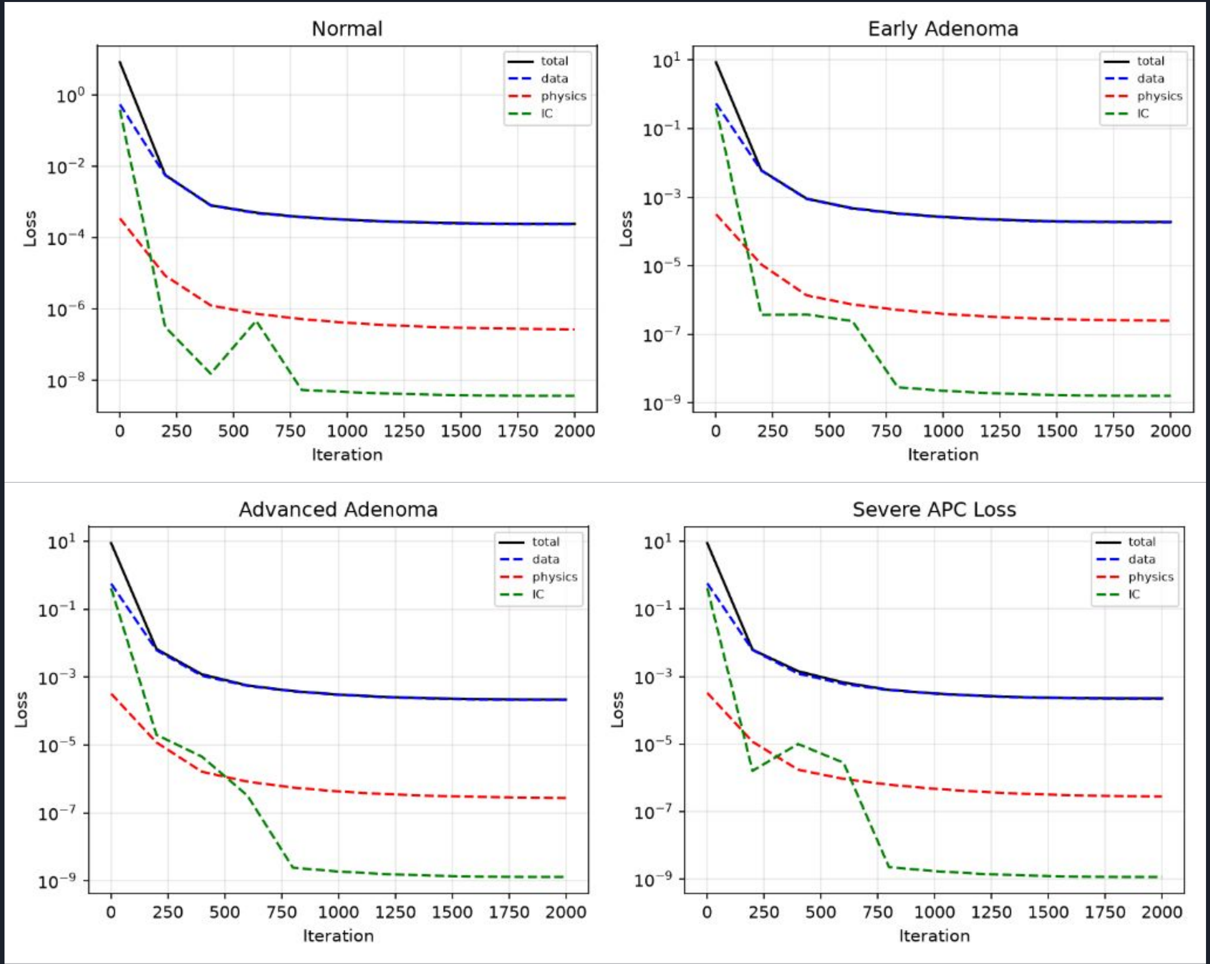
Forward PINN Loss



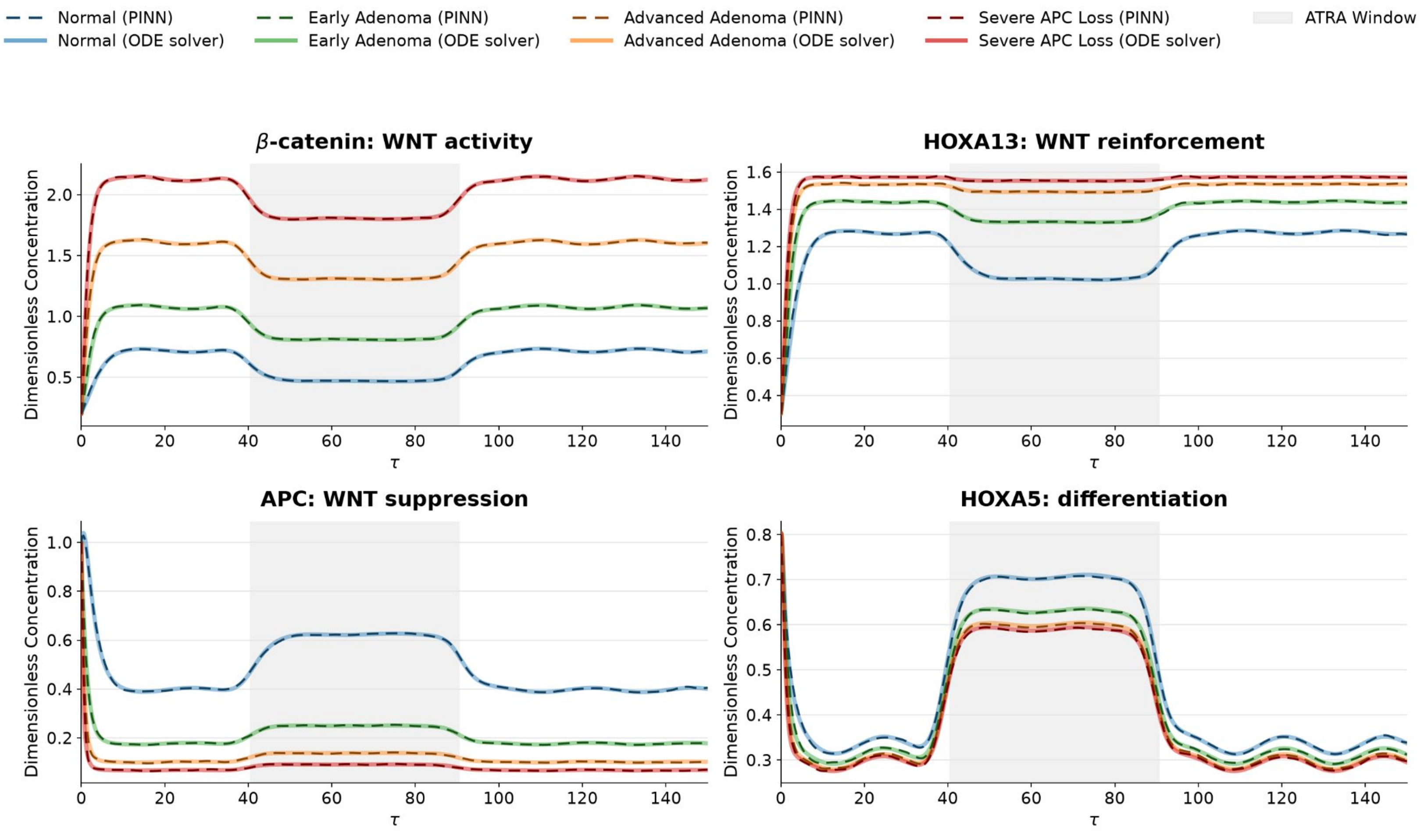

Core WNT-HOX Regulatory Dynamics via Forward PINN
Normal (PINN)
Normal (ODE solver)
Early Adenoma (PINN)
Early Adenoma (ODE solver)
Advanced Adenoma (PINN)
Advanced Adenoma (ODE solver)
Severe APC Loss (PINN)
Severe APC Loss (ODE solver)
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ

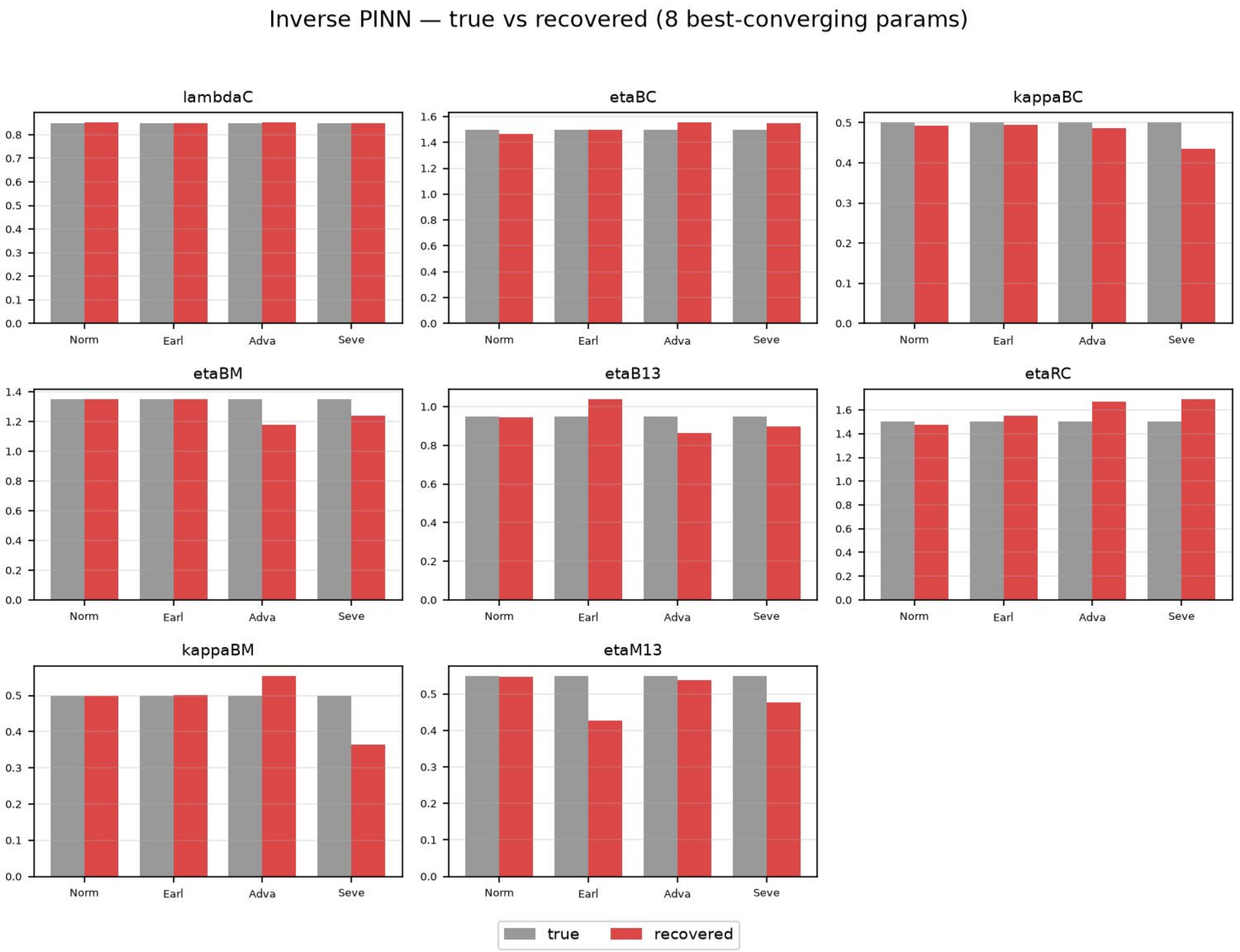

Inverse PINN
NN
Hidden Layers
Input
Output
t
σ
$\hat{x}$
DE
parameters
$\frac{d\hat{x}_1}{dt}$
$\frac{d\hat{x}_n}{dt}$
θ
DE
$\mathcal{L}_{\text{data}}$
$\mathcal{L}_{\text{ic}}$
$\mathcal{L}_{\text{phys}}$
Loss
$< \epsilon\,?$
YES
Done
NO

Inverse PINN Loss

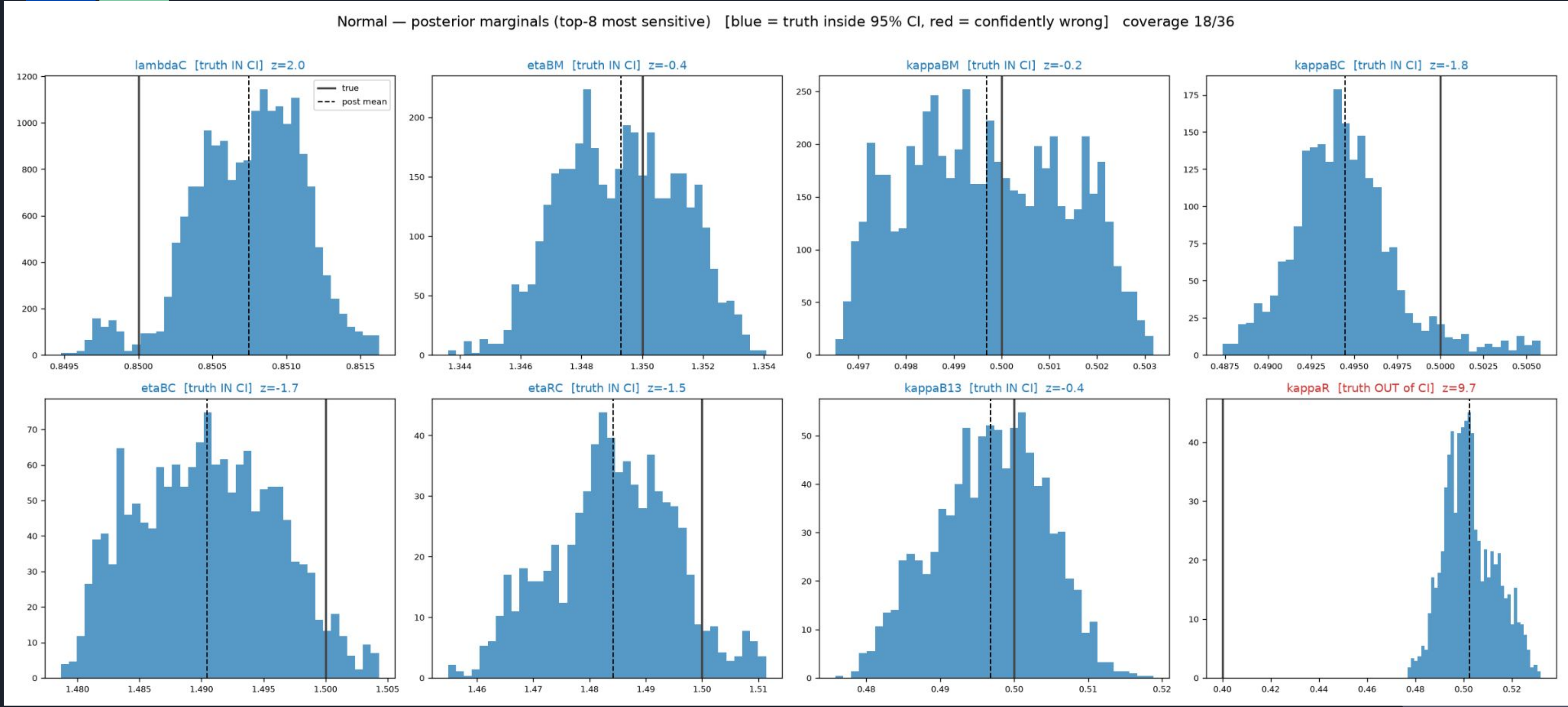


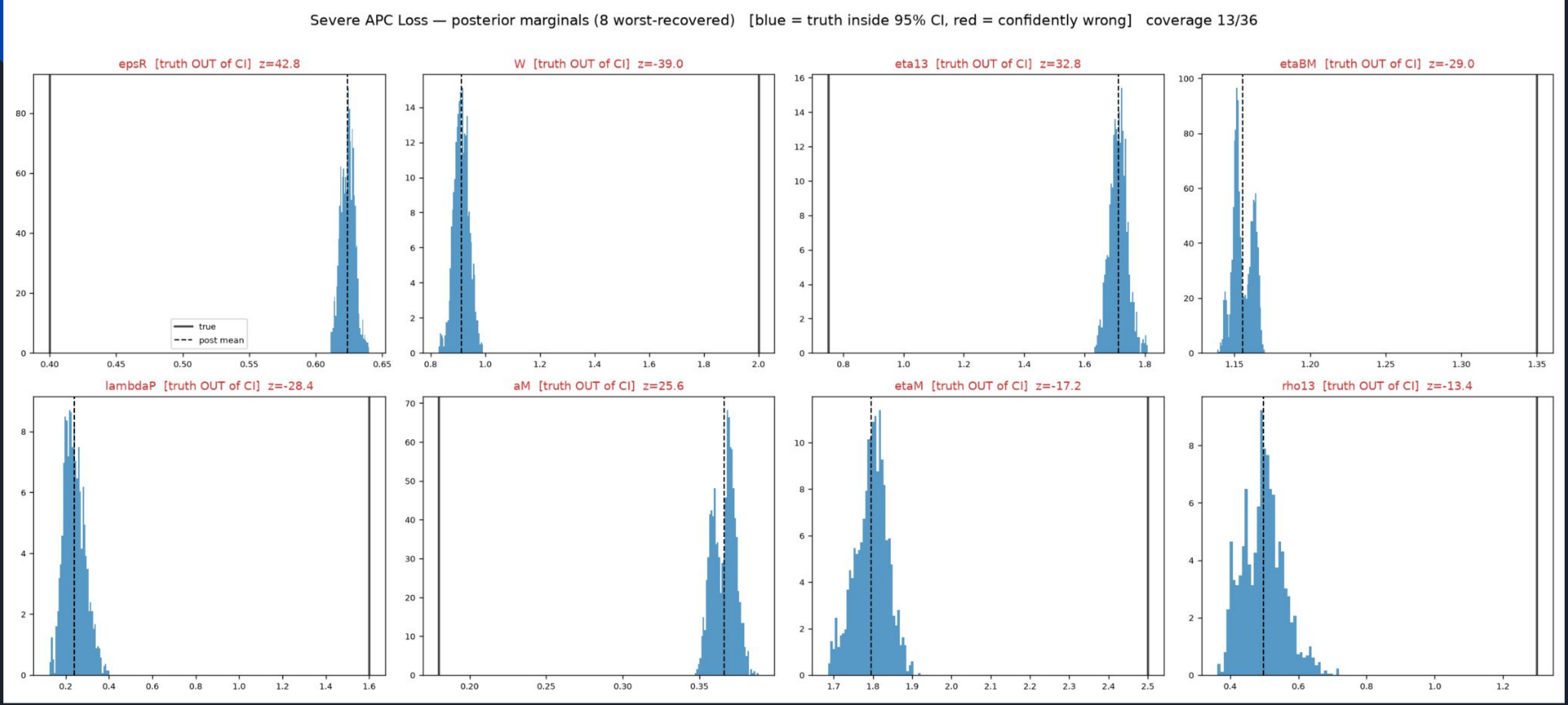
Core WNT-HOX Regulatory Dynamics via Inverse PINN




Inverse PINN — true vs recovered (8 best-converging params)
lambdaC
etaBC
kappaBC
etaBM
etaB13
etaRC
kappaBM
etaM13
Norm
Earl
Adva
Seve
true
recovered

Inverse Bayesian PINN - uncertainty quantification




Severe APC Loss — posterior marginals (8 worst-recovered) [blue = truth inside 95% CI, red = confidently wrong] coverage 13/36
epsR [truth OUT of CI] z=42.8
W [truth OUT of CI] z=-39.0
eta13 [truth OUT of CI] z=32.8
etaBM [truth OUT of CI] z=-29.0
lambdaP [truth OUT of CI] z=-28.4
aM [truth OUT of CI] z=25.6
etaM [truth OUT of CI] z=-17.2
rho13 [truth OUT of CI] z=-13.4
true
post mean

Parameter Recovery Comparison

	Bayesian inverse PINN	**Inverse PINN**	**ODE-fit**
“Normal”	**19**	**17**	**18/36**
“Early Adenoma”	**20**	**16**	**17/36**
“Advanced Adenoma”	**11**	**10**	**13/36**
“Severe APC Loss”	**8**	**7**	**11/36**

Next Steps

- Improve Bayesian PINN
    - Investigate limitations
- Hybrid Mechanistic-Neural Model
- Stochastic Modeling

Key References

1. **Raissi, Perdikaris & Karniadakis (2019).** Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear PDEs. *J. Comput. Phys.* **378**, 686–707.

   Foundational PINN formulation — the data + physics-residual loss used in every model here.

2. **Yazdani, Lu, Raissi & Karniadakis (2020).** Systems biology informed deep learning for inferring parameters and hidden dynamics. *PLoS Comput. Biol.* **16**(11), e1007575.

   Closest template to this work: inverse PINNs on biological ODE networks; the local-sensitivity/Fisher route to identifiability.

3. **Yang, Meng & Karniadakis (2021).** B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *J. Comput. Phys.* **425**, 109913.

   Basis of the Bayesian inverse PINN: HMC over parameters gives posterior marginals instead of a point estimate.

4. **Raue *et al.* (2009).** Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics* **25**(15), 1923–1929.

   The identifiability verdict — profile likelihood distinguishes the structural from the practical (high-WNT θ_P) wall.

5. **Wang, Yu & Perdikaris (2022).** When and why PINNs fail to train: a neural tangent kernel perspective. *J. Comput. Phys.* **449**, 110768.

   Explains the loss-term imbalance and spectral bias behind the training pathologies; motivates adaptive loss weighting.

6. **Tancik *et al.* (2020).** Fourier features let networks learn high frequency functions in low dimensional domains. *NeurIPS* **33**, 7537–7547.

   Fourier-feature time embedding — required to capture the circadian RA forcing and the ATRA pulse.

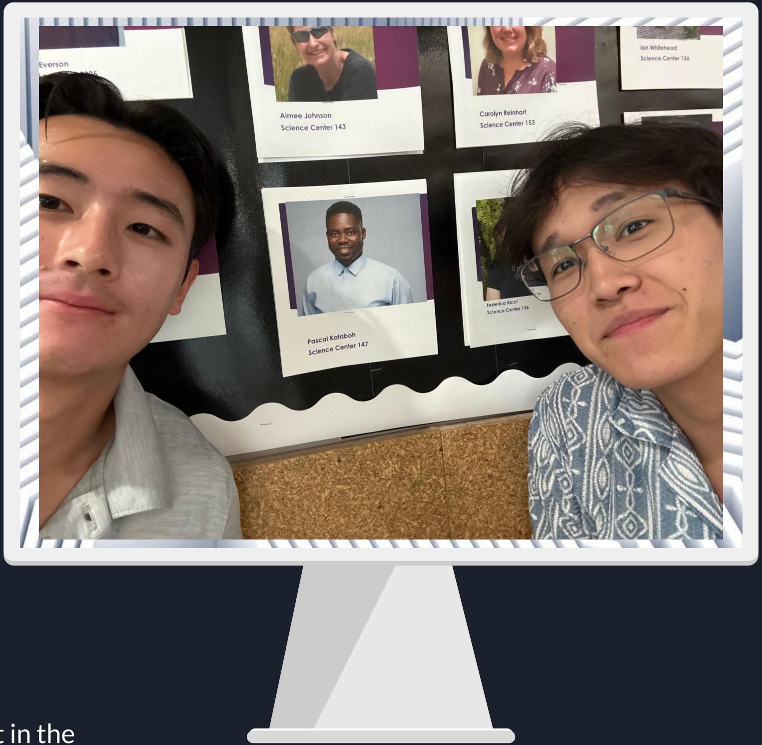
7. **Ji, Qiu, Zhang & Deng (2021).** Stiff-PINN: physics-informed neural network for stiff chemical kinetics. *J. Phys. Chem. A* **125**(36), 8098–8106.

   Diagnoses PINN failure on stiff multiscale kinetics — the same stiffness that shapes the WNT–RA–HOX residual.

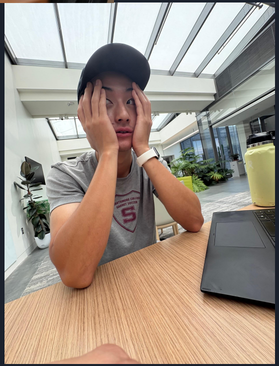
Thank you!

Questions? (we will try to answer😅)

PK calling Amir + Nate to check in on our progress…



4th diet coke of the day + Celsius b/c his sleep schedule is so cooked



Don't know what in the world is going on b/c he spent more hours watching the world cup than working

Thank you very much to the Provost's Office for funding our summer research project!